Storage Lab: Enterprise LVM Volume Extension Under Live I/O Load with Rollback SOP

Duration : 90 Min.

Scenario

A production enterprise cluster volume /dev/vg_enterprise/lv_transactions (mounted at /opt/transactions) is at 96% capacity. An additional raw LUN /dev/sdc (100 GB) has been provisioned by the SAN administrator. The engineer must execute a zero-downtime volume extension: initialize /dev/sdc as a Physical Volume, add it to Volume Group vg_enterprise, extend the Logical Volume lv_transactions by 50 GB, expand the filesystem online, leave the remaining 50 GB in the VG as an emergency reserve, and formulate a rollback validation procedure.

Task Objectives

Perform the following enterprise LVM volume extension tasks:

> **Pre-Staged Files in `student_workspace/`:** - **`lvm_extension_signoff.json`**: Starter change-control signoff template.

1. Capture baseline LVM topology and record the pre-change state of volume group `vg_enterprise`.
2. Initialize the provisioned raw LUN `/dev/sdc` and extend the volume group.
3. Extend the logical volume `lv_transactions` by 50 GB, preserving remaining unallocated capacity as an emergency reserve.
4. Expand the live filesystem online under continuous I/O load without data corruption or transaction errors.
5. Document change-control verification details and volume attributes in `lvm_extension_signoff.json`.

Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.
2. Execute the required storage management commands, volume operations, or configuration edits.
3. If configuration files, backup manifests, or diagnostic reports are required, save them inside the `student_workspace` directory.
4. Verify your storage configurations locally by executing: `bash python run.py`
5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Pre-change capture: baseline LVM topology and mount states accurately recorded in lvm_before.txt	3 Marks
TC2	Physical volume creation: pvcreate successfully initializes /dev/sdc with proper partition alignment	3 Marks
TC3	Volume group expansion: vgextend incorporates /dev/sdc into vg_enterprise without disruptive locking	3 Marks
TC4	Targeted LV growth: lvextend extends logical volume by exactly +50GB rather than consuming all space	3 Marks
TC5	Filesystem expansion: filesystem grown online while continuous background write script runs without I/O error	3 Marks
TC6	Emergency headroom: verification that volume group maintains unallocated extents for future emergency growth	3 Marks
TC7	Change sign-off: lvm_extension_signoff.json documents before/after sizes, UUIDs, and verification checksums	2 Marks

Total Score: 20 Marks

Important Notes

- This is an **Auto-Evaluated** storage assessment.
- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.